

Average Causal Effect in Observational Studies II

Zhichao Jiang

Sun Yat-sen University

Spring 2026

Other estimation strategies

- Other use of propensity score in estimating ACE
- Calibration methods
- Matching

Regression weighted by inverse of propensity score

- Hajek estimator

$$\widehat{ACE}^{\text{hajek}} = \frac{\sum_{i=1}^n \frac{Z_i Y_i}{\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{Z_i}{\widehat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^n \frac{(1-Z_i) Y_i}{1-\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)}}$$

Regression weighted by inverse of propensity score

- Hajek estimator

$$\widehat{ACE}^{\text{hajek}} = \frac{\sum_{i=1}^n \frac{Z_i Y_i}{\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{Z_i}{\widehat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^n \frac{(1-Z_i) Y_i}{1-\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)}}$$

- Weighted least squares estimation

$$\begin{aligned} (\widehat{\beta}_0^{\text{wls}}, \widehat{\beta}_1^{\text{wls}}) &= \underset{\beta_0, \beta_1}{\operatorname{argmin}} \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 Z_i)^2 \\ w_i &= \frac{Z_i}{\widehat{e}(\mathbf{X}_i)} + \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)} = \begin{cases} \frac{Z_i}{\widehat{e}(\mathbf{X}_i)} & \text{if } Z_i = 1 \\ \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)} & \text{if } Z_i = 0 \end{cases} \end{aligned}$$

Regression weighted by inverse of propensity score

- Hajek estimator

$$\widehat{ACE}^{\text{hajek}} = \frac{\sum_{i=1}^n \frac{Z_i Y_i}{\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{Z_i}{\widehat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^n \frac{(1-Z_i) Y_i}{1-\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)}}$$

- Weighted least squares estimation

$$\begin{aligned} (\widehat{\beta}_0^{\text{wls}}, \widehat{\beta}_1^{\text{wls}}) &= \underset{\beta_0, \beta_1}{\operatorname{argmin}} \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 Z_i)^2 \\ w_i &= \frac{Z_i}{\widehat{e}(\mathbf{X}_i)} + \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)} = \begin{cases} \frac{Z_i}{\widehat{e}(\mathbf{X}_i)} & \text{if } Z_i = 1 \\ \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)} & \text{if } Z_i = 0 \end{cases} \end{aligned}$$

- Variance estimation: bootstrap

Weighted regression with covariates

- In randomized experiments, we run $\text{lm}(Y_i \sim 1 + Z_i + X_i + Z_i X_i)$ to improve efficiency

Weighted regression with covariates

- In randomized experiments, we run $\text{lm}(Y_i \sim 1 + Z_i + X_i + Z_i X_i)$ to improve efficiency
- In observational studies, can we run $\text{lm}(Y_i \sim 1 + Z_i + X_i + Z_i X_i)$ with weights
 - equivalent to two regressions: $\mathbb{E}(Y_i \mid Z_i = z, X_i) = \alpha_z + \gamma_z^\top X_i$
 - ACE estimator: $\hat{\alpha}_1^{\text{wls}} - \hat{\alpha}_0^{\text{wls}}$

Weighted regression with covariates

- Consider a doubly robust estimator based on linear models

$$\mathbb{E}(Y_i \mid Z_i = z, \mathbf{X}_i) = \alpha_z + \gamma_z^\top \mathbf{X}_i$$

$$\hat{\mu}_{1,\text{DR}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{Z_i \{Y_i - \mu_1(\mathbf{X}_i, \hat{\alpha}_1^{\text{wls}}, \hat{\gamma}_1^{\text{wls}})\}}{e(\mathbf{X}_i, \hat{\beta})} + \mu_1(\mathbf{X}_i, \hat{\alpha}_1^{\text{wls}}, \hat{\gamma}_1^{\text{wls}}) \right]$$

$$\hat{\mu}_{0,\text{DR}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{(1 - Z_i) \{Y_i - \mu_0(\mathbf{X}_i, \hat{\alpha}_0^{\text{wls}}, \hat{\gamma}_0^{\text{wls}})\}}{1 - e(\mathbf{X}_i, \hat{\beta})} + \mu_0(\mathbf{X}_i, \hat{\alpha}_0^{\text{wls}}, \hat{\gamma}_0^{\text{wls}}) \right]$$

$$\widehat{\text{ACE}}_{\text{DR}} = \hat{\mu}_{1,\text{DR}} - \hat{\mu}_{0,\text{DR}}$$

Weighted regression with covariates

- Consider a doubly robust estimator based on linear models

$$\mathbb{E}(Y_i \mid Z_i = z, \mathbf{X}_i) = \alpha_z + \gamma_z^\top \mathbf{X}_i$$

$$\hat{\mu}_{1,\text{DR}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{Z_i \{Y_i - \mu_1(\mathbf{X}_i, \hat{\alpha}_1^{\text{wls}}, \hat{\gamma}_1^{\text{wls}})\}}{e(\mathbf{X}_i, \hat{\beta})} + \mu_1(\mathbf{X}_i, \hat{\alpha}_1^{\text{wls}}, \hat{\gamma}_1^{\text{wls}}) \right]$$

$$\hat{\mu}_{0,\text{DR}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{(1 - Z_i) \{Y_i - \mu_0(\mathbf{X}_i, \hat{\alpha}_0^{\text{wls}}, \hat{\gamma}_0^{\text{wls}})\}}{1 - e(\mathbf{X}_i, \hat{\beta})} + \mu_0(\mathbf{X}_i, \hat{\alpha}_0^{\text{wls}}, \hat{\gamma}_0^{\text{wls}}) \right]$$

$$\widehat{\text{ACE}}_{\text{DR}} = \hat{\mu}_{1,\text{DR}} - \hat{\mu}_{0,\text{DR}}$$

- From the property of weighted least squares

$$\widehat{\text{ACE}}_{\text{DR}} = \hat{\alpha}_1^{\text{wls}} - \hat{\alpha}_0^{\text{wls}} + \bar{\mathbf{X}}^\top (\hat{\gamma}_1^{\text{wls}} - \hat{\gamma}_0^{\text{wls}})$$

Structural model

- Model $Y_i(z)$ instead of Y_i

Structural model

- Model $Y_i(z)$ instead of Y_i
 - Estimation involves $Y_i(z)$, e.g., MLE, moment estimation

Structural model

- Model $Y_i(z)$ instead of Y_i
 - Estimation involves $Y_i(z)$, e.g., MLE, moment estimation
- Under ignorability, for any function h

$$\mathbb{E}\{h(Y_i(1), X_i)\} = \mathbb{E}\left\{\frac{Z_i h(Y_i, X_i)}{e(X_i)}\right\}$$

$$\mathbb{E}\{h(Y_i(0), X_i)\} = \mathbb{E}\left\{\frac{(1 - Z_i) h(Y_i, X_i)}{1 - e(X_i)}\right\}$$

Structural model

- Model $Y_i(z)$ instead of Y_i
 - Estimation involves $Y_i(z)$, e.g., MLE, moment estimation
- Under ignorability, for any function h

$$\mathbb{E}\{h(Y_i(1), X_i)\} = \mathbb{E}\left\{\frac{Z_i h(Y_i, X_i)}{e(X_i)}\right\}$$

$$\mathbb{E}\{h(Y_i(0), X_i)\} = \mathbb{E}\left\{\frac{(1 - Z_i) h(Y_i, X_i)}{1 - e(X_i)}\right\}$$

- Estimation involves only Y_i with inverse propensity score weighting

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$
 - include propensity score (spline basis) in the outcome model (Little and An, 2004; Zhang and Little, 2009)
 - include $1/e(X)$ as regressor in the outcome model (Scharfstein, 1997; Bang and Robins, 2005)

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$
 - include propensity score (spline basis) in the outcome model (Little and An, 2004; Zhang and Little, 2009)
 - include $1/e(X)$ as regressor in the outcome model (Scharfstein, 1997; Bang and Robins, 2005)
- Procedure for estimating $\mathbb{E}\{Y_i(1)\}$:

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$
 - include propensity score (spline basis) in the outcome model (Little and An, 2004; Zhang and Little, 2009)
 - include $1/e(X)$ as regressor in the outcome model (Scharfstein, 1997; Bang and Robins, 2005)
- Procedure for estimating $\mathbb{E}\{Y_i(1)\}$:
 - fit a model for propensity score $\text{logit}\{e(X_i)\} = X_i^\top \beta$, obtain $\hat{e}(X_i)$

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$
 - include propensity score (spline basis) in the outcome model (Little and An, 2004; Zhang and Little, 2009)
 - include $1/e(X)$ as regressor in the outcome model (Scharfstein, 1997; Bang and Robins, 2005)
- Procedure for estimating $\mathbb{E}\{Y_i(1)\}$:
 - fit a model for propensity score $\text{logit}\{e(X_i)\} = X_i^\top \beta$, obtain $\hat{e}(X_i)$
 - fit a linear regression model of $Y_i(1)$ on X_i and $1/\hat{e}(X_i)$
 $\mathbb{E}(Y_i(1) \mid X_i) = X_i^\top \gamma + \phi/\hat{e}(X_i)$, obtain $\hat{\gamma}$ and $\hat{\phi}$

Regression with propensity score as a covariate

- Under ignorability, $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(X_i)$
 - include propensity score (spline basis) in the outcome model (Little and An, 2004; Zhang and Little, 2009)
 - include $1/e(X)$ as regressor in the outcome model (Scharfstein, 1997; Bang and Robins, 2005)
- Procedure for estimating $\mathbb{E}\{Y_i(1)\}$:
 - fit a model for propensity score $\text{logit}\{e(X_i)\} = X_i^\top \beta$, obtain $\hat{e}(X_i)$
 - fit a linear regression model of $Y_i(1)$ on X_i and $1/\hat{e}(X_i)$
 $\mathbb{E}(Y_i(1) \mid X_i) = X_i^\top \gamma + \phi/\hat{e}(X_i)$, obtain $\hat{\gamma}$ and $\hat{\phi}$
 - calculate the estimator for $\mathbb{E}\{Y_i(1)\}$

$$\hat{\mu}_R = \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}\{Y(1) \mid X_i\} = \frac{1}{n} \sum_{i=1}^n \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right)$$

Double robustness

- From the property of linear regression

$$\frac{1}{n} \sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)} \left\{ Y_i - \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) \right\} = 0 \quad (1)$$

Double robustness

- From the property of linear regression

$$\frac{1}{n} \sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)} \left\{ Y_i - \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) \right\} = 0 \quad (1)$$

- $\hat{\mu}_R$ has the same expression as the doubly robust estimator

$$\hat{\mu}_R = \frac{1}{n} \sum_{i=1}^n \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) + \sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)} \left\{ Y_i - \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) \right\}$$

Double robustness

- From the property of linear regression

$$\frac{1}{n} \sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)} \left\{ Y_i - \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) \right\} = 0 \quad (1)$$

- $\hat{\mu}_R$ has the same expression as the doubly robust estimator

$$\hat{\mu}_R = \frac{1}{n} \sum_{i=1}^n \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) + \sum_{i=1}^n \frac{Z_i}{\hat{e}(X_i)} \left\{ Y_i - \left(X_i^\top \hat{\gamma} + \frac{\hat{\phi}}{\hat{e}(X_i)} \right) \right\}$$

- $\hat{\mu}_R$ is consistent if either outcome or propensity score model is correctly specified

Generalization

- Regression estimator becomes doubly robust estimator under Equation (1)

Generalization

- Regression estimator becomes doubly robust estimator under Equation (1)
 - linear model with $1/\hat{e}(X_i)$ as regressor guarantees Equation (1)

Generalization

- Regression estimator becomes doubly robust estimator under Equation (1)
 - linear model with $1/\hat{e}(X_i)$ as regressor guarantees Equation (1)
 - $X_i^\top \gamma$ determines the outcome model

Generalization

- Regression estimator becomes doubly robust estimator under Equation (1)
 - linear model with $1/\hat{e}(X_i)$ as regressor guarantees Equation (1)
 - $X_i^\top \gamma$ determines the outcome model
- Generalization: $\mathbb{E}\{Y_i(1) \mid X_i\} = \hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\} + \phi/\hat{e}(X_i)$, where $\hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\}$ is an outcome regression estimator

Generalization

- Regression estimator becomes doubly robust estimator under Equation (1)
 - linear model with $1/\hat{e}(X_i)$ as regressor guarantees Equation (1)
 - $X_i^\top \gamma$ determines the outcome model
- Generalization: $\mathbb{E}\{Y_i(1) \mid X_i\} = \hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\} + \phi/\hat{e}(X_i)$, where $\hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\}$ is an outcome regression estimator
- $\hat{\mu}_R$ is consistent if either $\hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\}$ or $\hat{e}(X_i)$ is consistent

Targeted maximum likelihood estimation (TMLE)

- Estimate $\mathbb{E}\{Y_i(1) \mid X_i\}$ to get initial estimator $\hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\}$
- Estimate propensity score to get $\hat{e}(X_i)$
- Construct a “clever” covariate: $1/\hat{e}(X_i)$
- Update the outcome model to $\hat{\mathbb{E}}^1(Y(1) \mid X)$ by fitting $\mathbb{E}\{Y_i(1) \mid X_i\} = \hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\} + \phi/\hat{e}(X_i)$
- Estimator: $\frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}^1\{Y_i(1) \mid X_i\}$

Targeted maximum likelihood estimation (TMLE)

- Estimate $\mathbb{E}\{Y_i(1) \mid X_i\}$ to get initial estimator $\hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\}$
- Estimate propensity score to get $\hat{e}(X_i)$
- Construct a “clever” covariate: $1/\hat{e}(X_i)$
- Update the outcome model to $\hat{\mathbb{E}}^1(Y(1) \mid X)$ by fitting $\mathbb{E}\{Y_i(1) \mid X_i\} = \hat{\mathbb{E}}^0\{Y_i(1) \mid X_i\} + \phi/\hat{e}(X_i)$
- Estimator: $\frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}^1\{Y_i(1) \mid X_i\}$

- Van der laan and Rubin (2006) propose TMLE
 - similar procedure for ACE
 - link function can be used for different types of outcome (e.g., logit link for binary outcome)

Calibration methods

- Balancing property of $e(X_i)$

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model
 - weights w_i should satisfy balancing property

$$\mathbb{E}\{w_i Z_i h(X_i)\} = \mathbb{E}\{w_i (1 - Z_i) h(X_i)\} = \mathbb{E}\{h(X_i)\}$$

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model
 - weights w_i should satisfy balancing property

$$\mathbb{E}\{w_i Z_i h(X_i)\} = \mathbb{E}\{w_i (1 - Z_i) h(X_i)\} = \mathbb{E}\{h(X_i)\}$$

- ACE estimation

$$\text{ACE} = \mathbb{E}\{w_i Z_i Y_i\} - \mathbb{E}\{w_i (1 - Z_i) h(X_i)\}$$

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model
 - weights w_i should satisfy balancing property

$$\mathbb{E}\{w_i Z_i h(X_i)\} = \mathbb{E}\{w_i (1 - Z_i) h(X_i)\} = \mathbb{E}\{h(X_i)\}$$

- ACE estimation

$$ACE = \mathbb{E}\{w_i Z_i Y_i\} - \mathbb{E}\{w_i (1 - Z_i) h(X_i)\}$$

- Weights are not unique $\rightsquigarrow$ minimize some functions of the weights

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model
 - weights w_i should satisfy balancing property

$$\mathbb{E}\{w_i Z_i h(X_i)\} = \mathbb{E}\{w_i (1 - Z_i) h(X_i)\} = \mathbb{E}\{h(X_i)\}$$

- ACE estimation

$$ACE = \mathbb{E}\{w_i Z_i Y_i\} - \mathbb{E}\{w_i (1 - Z_i) h(X_i)\}$$

- Weights are not unique $\rightsquigarrow$ minimize some functions of the weights
 - e.g., weights to minimize the variance of the ACE estimator

Calibration methods

- Balancing property of $e(X_i)$
- Directly estimate the weights w_i without imposing model
 - weights w_i should satisfy balancing property

$$\mathbb{E}\{w_i Z_i h(X_i)\} = \mathbb{E}\{w_i (1 - Z_i) h(X_i)\} = \mathbb{E}\{h(X_i)\}$$

- ACE estimation

$$ACE = \mathbb{E}\{w_i Z_i Y_i\} - \mathbb{E}\{w_i (1 - Z_i) h(X_i)\}$$

- Weights are not unique $\rightsquigarrow$ minimize some functions of the weights
 - e.g., weights to minimize the variance of the ACE estimator
 - implementation using optimization

Calibration Methods

- Entropy balancing (Hainmueller. 2012. *Political Anal.*)

$$\{w_1^*, w_2^*, \dots, w_{n_0}^*\} = \underset{w}{\operatorname{argmin}} \sum_{i:Z_i=0} w_i \log(w_i/q_i)$$

subject to

$$w_i \geq 0, \quad \sum_{i:Z_i=0} w_i = 1, \quad \sum_{i:Z_i=0} w_i f(\mathbf{x}_i) = \frac{1}{n} \sum_i f(\mathbf{x}_i)$$

- Stable weights (Zubizarreta, 2015, *JASA*)

$$\{w_1^*, w_2^*, \dots, w_n^*\} = \underset{w}{\operatorname{argmin}} \|\mathbf{w} - \bar{\mathbf{w}}\|_2^2$$

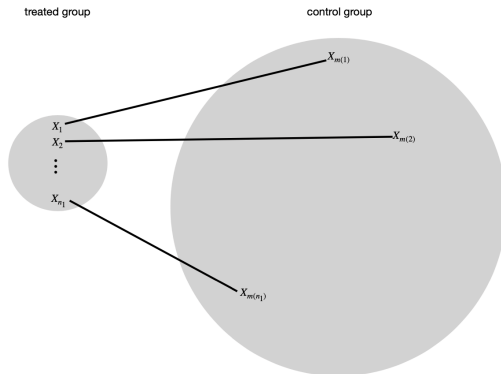
subject to

$$w_i \geq 0, \quad \sum_{i:Z_i=0} w_i = 1, \quad \left| \sum_{i:Z_i=0} w_i X_{ij} - \frac{1}{n} \sum_i X_{ij} \right| \leq \delta_j$$

Matching

- Popular in empirical research
- Easy implementation without modeling
- Complicated theory
- A practical guide for matching: Stuart (2010) “Matching methods for causal inference: A review and a look forward”

Exact matching



- Exact matching $\rightsquigarrow$ perfect covariate balance
- No model dependence
- Infeasible when there are many covariates and covariates are continuous

Matching based on distance measures

- Matching based on distances between covariates $d(\mathbf{X}_i, \mathbf{X}_j)$
 - Euclidean distance $d(\mathbf{X}_i, \mathbf{X}_j) = \|\mathbf{X}_i - \mathbf{X}_j\|_2^2$
 - Mahalanobis distance $d(\mathbf{X}_i, \mathbf{X}_j) = (\mathbf{X}_i - \mathbf{X}_j)^\top \Omega^{-1}(\mathbf{X}_i - \mathbf{X}_j)$
- Different types of matching
 - One to one, one to many
 - With or without replacement
 - Optimal matching
 - Full matching
- For a given unit in the treatment (control) group, find the unit with the smallest distance in the control (treatment) group
 - 1 to M matching ($M > 1$)
 - depends on the distance measure
 - may drop units that are hard to find matches
 - adjustment after matching

Matching estimator for the ACE

- For a treated unit i , we find the M matched units in the control group and denote $\tilde{Y}_i = \frac{1}{M} \sum_{k \in \mathcal{M}_i} Y_k$, where $\mathcal{M}_i$ is the set of matched units from the control group for unit i
- For a control unit i , we find the M matched units in the treatment group and denote $\tilde{Y}_i = \frac{1}{M} \sum_{k \in \mathcal{M}_i} Y_k$, where $\mathcal{M}_i$ is the set of matched units from the treatment group for unit i

Matching estimator for the ACE

- For a treated unit i , we find the M matched units in the control group and denote $\tilde{Y}_i = \frac{1}{M} \sum_{k \in \mathcal{M}_i} Y_k$, where $\mathcal{M}_i$ is the set of matched units from the control group for unit i
- For a control unit i , we find the M matched units in the treatment group and denote $\tilde{Y}_i = \frac{1}{M} \sum_{k \in \mathcal{M}_i} Y_k$, where $\mathcal{M}_i$ is the set of matched units from the treatment group for unit i
- Matching estimator

$$\widehat{\text{ACE}}^{\text{matching}} = \frac{1}{n} \left\{ \sum_{i: Z_i=1} (Y_i - \tilde{Y}_i) + \sum_{i: Z_i=0} (\tilde{Y}_i - Y_i) \right\}$$

Matching estimator for the ACE

- Abadie and Imbens (2006, 2008, 2011) study the properties of the matching estimator with replacement assuming $(Z_i, X_i, Y_i(1), Y_i(0))$ are i.i.d.

Matching estimator for the ACE

- Abadie and Imbens (2006, 2008, 2011) study the properties of the matching estimator with replacement assuming $(Z_i, X_i, Y_i(1), Y_i(0))$ are i.i.d.
- $\widehat{ACE}^{\text{matching}}$ has non-negligible bias especially when X is multidimensional

Matching estimator for the ACE

- Abadie and Imbens (2006, 2008, 2011) study the properties of the matching estimator with replacement assuming $(Z_i, X_i, Y_i(1), Y_i(0))$ are i.i.d.
- $\widehat{ACE}^{\text{matching}}$ has non-negligible bias especially when X is multidimensional
- Estimator for the bias $\hat{B} = n^{-1} \sum_{i=1}^n \hat{B}_i$

$$\hat{B}_i = \begin{cases} M^{-1} \sum_{k \in \mathcal{M}_i} \{\hat{\mu}_0(X_i) - \hat{\mu}_0(X_k)\} & Z_i = 1 \\ M^{-1} \sum_{k \in \mathcal{M}_i} \{\hat{\mu}_1(X_k) - \hat{\mu}_1(X_i)\} & Z_i = 0 \end{cases}$$

- $\{\hat{\mu}_1(X_i), \hat{\mu}_0(X_i)\}$ are the predicted outcomes, e.g., from OLS
- bias corrected estimator: $\widehat{ACE}^{\text{mbc}} = \widehat{ACE}^{\text{matching}} - \hat{B}$

Matching estimator for the ACE

- Linear expansion form: $\widehat{ACE}^{mbc} = n^{-1} \sum_{i=1}^n \hat{\psi}_i$

$$\hat{\psi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + (2Z_i - 1)(1 + K_i/M) \{Y_i - \hat{\mu}_{Z_i}(X_i)\}$$

- K_i is the number of times that unit i is used in a match

Matching estimator for the ACE

- Linear expansion form: $\widehat{ACE}^{mbc} = n^{-1} \sum_{i=1}^n \hat{\psi}_i$

$$\hat{\psi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + (2Z_i - 1)(1 + K_i/M) \{Y_i - \hat{\mu}_{Z_i}(X_i)\}$$

- K_i is the number of times that unit i is used in a match
- Variance estimator

$$\widehat{\text{var}}\{\widehat{ACE}^{mbc}\} = \frac{1}{n^2} \sum_{i=1}^n (\hat{\psi}_i - \widehat{ACE}^{mbc})^2$$

Matching estimator for the ACE

- Linear expansion form: $\widehat{ACE}^{mbc} = n^{-1} \sum_{i=1}^n \hat{\psi}_i$

$$\hat{\psi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + (2Z_i - 1)(1 + K_i/M) \{Y_i - \hat{\mu}_{Z_i}(X_i)\}$$

- K_i is the number of times that unit i is used in a match

- Variance estimator

$$\widehat{\text{var}}\{\widehat{ACE}^{mbc}\} = \frac{1}{n^2} \sum_{i=1}^n (\hat{\psi}_i - \widehat{ACE}^{mbc})^2$$

- Otsu and Rai (2017) propose a bootstrap procedure based on the linear expansion

Connection with doubly robust estimator

- $\widehat{\text{ACE}}^{\text{mbc}}$ is equal to

$$\frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\} + \frac{1}{n} \sum_{i=1}^n \left\{ \left(1 + \frac{K_i}{M}\right) Z_i \hat{R}_i - \left(1 + \frac{K_i}{M}\right) (1 - Z_i) \hat{R}_i \right\}$$

Connection with doubly robust estimator

- $\widehat{ACE}^{mbc}$ is equal to

$$\frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\} + \frac{1}{n} \sum_{i=1}^n \left\{ \left(1 + \frac{K_i}{M}\right) Z_i \hat{R}_i - \left(1 + \frac{K_i}{M}\right) (1 - Z_i) \hat{R}_i \right\}$$

- $\hat{R}_i = Y_i - \hat{\mu}_{Z_i}(X_i)$ is the residual from outcome regression
- $n^{-1} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$ is the outcome regression estimator
- the second term is similar to the inverse probability weighting of residual R_i

Connection with doubly robust estimator

- $\widehat{ACE}^{\text{mbc}}$ is equal to

$$\frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\} + \frac{1}{n} \sum_{i=1}^n \left\{ \left(1 + \frac{K_i}{M}\right) Z_i \hat{R}_i - \left(1 + \frac{K_i}{M}\right) (1 - Z_i) \hat{R}_i \right\}$$

- $\hat{R}_i = Y_i - \hat{\mu}_{Z_i}(X_i)$ is the residual from outcome regression
 - $n^{-1} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$ is the outcome regression estimator
 - the second term is similar to the inverse probability weighting of residual R_i
- Matching can be viewed as a nonparametric method to estimate the propensity score
 - $1 + K_i/M$ should be similar to $1/\hat{e}(X_i)$ for $Z_i = 1$ and $1/\{1 - \hat{e}(X_i)\}$ for $Z_i = 0$
 - Lin et al. (2023) provide a formal theory

Summary

- The identification of the average treatment effect in observational studies typically requires:
 - 1 overlap
 - 2 ignorability
- Various methods to estimate causal effects:
 - use propensity score in regression
 - connection to DR estimator
 - Calibration methods
 - estimate weights without modeling
 - need to choose which covariates to balance
 - Matching
 - easy implementation but complicated theory
 - non-negligible bias $\rightsquigarrow$ bias corrected matching estimator
 - connection to weighting methods

Suggested readings

- Using propensity score in regression
 - DING. Chapter 14
 - Robins, 2007. “Comment: Performance of Double-Robust Estimators When “Inverse Probability” Weights Are Highly Variable”
- Calibration methods
 - papers by José Zubizarreta and references therein
- Matching
 - papers by Abadie and Imbens
 - Stuart, 2010. “Matching methods for causal inference: A review and a look forward”